

Problem

The result of $u(t) * u(t)$ is:

- (a) $u^2(t)$
- (b) $tu(t)$
- (c) $t^2u(t)$
- (d) $\delta(t)$

Step-by-step solution

Step 1 of 1

$$(b) \quad u(t) * u(t) = \int_{-\infty}^{\infty} y(\lambda) u(t-\lambda) d\lambda = \mathcal{L}^{-1} [F_1(S) F_2(S)]$$

$$u(t) * u(t) = \mathcal{L}^{-1} \left[\frac{1}{s^2} \right] = t u(t)$$